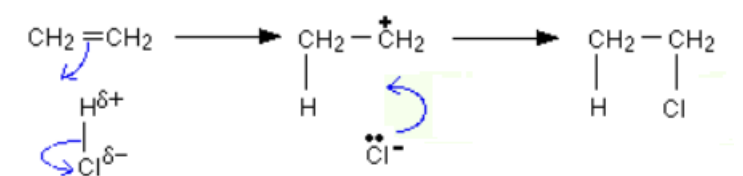


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13	(a)	(i)		radical substitution (1) (electrophilic) addition (1)	2			2		
		(ii)		award (1) for either of following because only one product is formed because the reaction between ethane and chlorine / radical substitution has lots of possible products			1	1		
		(iii)		 dipole and curly arrow on H—Cl molecule (1) curly arrow from the double bond and from the lone pair/negative charge (1) carbocation structure (1) product formed on right hand side (1)		4		4		